[7] Project Status and Pending Work Report

Project: Eventura (powered by Clientura) — Enterprise Event & Hackathon Platform
Document Version: 1.0.0
Current Milestone: Release Candidate 1.0 (v1.0.0 Production Ready)
Date: October 29, 2026
Audience: Product Managers, Engineering Leadership, Stakeholders, DevOps Team

1. Executive Status Summary

The **Eventura** software platform has successfully concluded its active development, optimization, security hardening, and pre-launch quality assurance cycles. The codebase is currently classified as **PRODUCTION READY (PRE-LAUNCH VERIFIED)**.

All core business requirements, user roles, security guardrails, and automated testing suites have achieved 100% compliance. The platform has been stress-tested across 5 concurrent event tracks, verified across 116 application routes, and compiled cleanly with zero TypeScript errors.

PRODUCTION READINESS SCORECARD

Capability / Area	Status	Confidence / Verification
System Architecture	100% Ready	Clean Layered Monorepo
Authentication & TOTP MFA	100% Ready	Universal Enforcement
Multi-Tenant Isolation	100% Ready	Zero Silent Fallbacks
Event & Proposal Workflows	100% Ready	Complete Lifecycle Coverage
Registrations & Payments	100% Ready	Stripe Integration Active
Submissions & View-Only Doc	100% Ready	Sandboxed Inline Streaming
Evaluation & Leaderboard	100% Ready	Rubric Scoring Verified
Attendance & QR Check-In	100% Ready	Signed QR Codes & Scanner
Certificates & Verification	100% Ready	Public Verification Engine
Reports & BI Dashboards	100% Ready	Full Aggregation & Export
Automated QA & Stress Test	100% Ready	116/116 Routes, 220 Ops

2. Completed Capabilities & Deliverables Breakdown

2.1 Multi-Tenant Organization Workspace

- Logical tenant container separating institutions, universities, and corporate accelerators.

- Subdomain / domain binding, custom branding configurations, and license tracking.
- Strict `x-organization-id` header validation with zero fallback to prevent cross-tenant exposure.

2.2 Role-Based Access Control (RBAC) & Security

- 8 Pre-seeded standard enterprise roles: `Sudo Admin`, `Organization Admin`, `Event Manager`, `Faculty Coordinator`, `Student Coordinator`, `Judge / Evaluator`, `Mentor`, `Participant`.
- Decoupled `RolePermission` matrix with 50+ granular capability toggles.
- Cryptographic password hashing (`bcrypt`) and universal Time-based One-Time Password (TOTP) MFA (`otplib`).

2.3 Hackathon Proposal & Event Management Lifecycle

- Structured proposal submission for student/faculty coordinators (budgeting, tracks, timelines).
- Manager review and approval engine that automatically spawns draft event records.
- Comprehensive event configurations: registration limits, team size bounds, venue/online modes, schedule tracks, and round criteria.

2.4 Attendee Registration & Stripe Payment Gateway

- Free and paid event registration tiers with coupon code discounts.
- Stripe Checkout Session creation and webhook handling (`checkout.session.completed`).
- Team creation, unique join-code distribution, and team leader role allocations.

2.5 Problem Statement Selection & Lock Rules

- Event-scoped problem statement directories with detailed briefs and evaluation criteria.
- Permanent selection lock once confirmed by team lead, preventing mid-competition statement swapping.

2.6 Deliverable Submission & Protected Document Streaming

- Project deliverables support GitHub URLs, live demonstration links, and uploaded files.
- Protected inline file viewing (`GET /api/v1/submissions/:id/files/:fileId/view`) verifying requester is a team member, assigned judge, or administrator before streaming.

2.7 Rubric Evaluation & Live Leaderboards

- Quantitative scorecard grading against configured rubric criteria (e.g. Innovation, Technical Depth).
- Score validation against `maxScore` constraints.
- Real-time leaderboard aggregation ranking teams deterministically.

2.8 QR Code Attendance Engine

- Generation of HMAC-signed QR passes for registered participants.
- Integrated camera scanner (`html5-qrcode`) for on-site gate check-ins with anti-duplicate validation.

2.9 Cryptographic Certificates & Verification

- Automated vector PDF certificate generation with embedded verification QR code.
- Public URL route (`/certificates/verify/:code`) enabling instant third-party authenticity verification.

2.10 Business Intelligence & Analytics Hub

- Executive dashboards aggregating registrations, revenue MRR/MTD, attendance percentages, and score distributions.
- CSV and PDF report export engines.
- Extended dashboards for Recruitment, Sponsors, Community, Learning, and Feedback surveys.

---

3. Remediated Defects & Quality Hardening

During the pre-launch stabilization phase, the engineering team resolved all critical and high-priority findings:

1. **Eliminated Administrative MFA Bypass**: Audited and removed hardcoded condition in `auth.service.ts`. MFA is universally enforced.
2. **Eliminated Silent Tenant Fallback**: Replaced automatic database fallback in `tenant.middleware.ts` with strict HTTP 400 rejection.
3. **Fixed Problem Statement Registration Status Check**: Expanded allowed statuses in `problem-statements.service.ts` to include `PAID`, `CONFIRMED`, and `REGISTERED`.
4. **Resolved Submission File 404s**: Implemented authorized `viewFile` endpoint with `Content-Disposition: inline` and absolute API URL resolution.
5. **Hardened Public Signup Form**: Restricted `/signup` strictly to `Participant` role cards, preventing unauthorized self-assignment of administrative roles.
6. **Synced Missing Reports Permissions**: Created idempotent database sync script backfilling `reports.read` and `reports.export` to manager/admin roles.

---

4. Pending Work & Future Product Roadmap

While the platform is 100% production-ready for launch, the following non-blocking architectural enhancements are planned for subsequent quarterly releases:

4.1 Phase 5: Cloud Object Storage Driver (Q1 2027)

- **Current State**: Deliverable files and generated certificates persist to a local volume mount.
- **Planned Enhancement**: Implement an S3-compatible cloud storage adapter (AWS S3, Cloudflare R2, or Google Cloud Storage) using presigned URLs.
- **Benefit**: Simplifies multi-region horizontal scaling of server pods without requiring shared NFS mounts.

4.2 Phase 6: Redis Pub/Sub WebSocket Leaderboard (Q1 2027)

- **Current State**: Live leaderboard updates utilize periodic TanStack Query polling (10-second intervals).
- **Planned Enhancement**: Deploy a lightweight WebSocket gateway backed by Redis Pub/Sub to push real-time score updates to stadium projection displays.
- **Benefit**: Eliminates HTTP polling overhead during mega-hackathons exceeding 10,000 live spectators.

4.3 Phase 7: Mobile PWA Native Push Notifications (Q2 2027)

- **Current State**: In-app notifications and email broadcasts via SMTP.
- **Planned Enhancement**: Integrate Web Push API with VAPID keys to send lock screen notifications to mobile devices when judging rounds start.
- **Benefit**: Increases participant engagement and check-in adherence during in-person events.

4.4 Phase 8: Multi-Currency & Automated Sales Tax in Stripe (Q2 2027)

- **Current State**: Single currency transactions (USD) with flat ticket pricing.
- **Planned Enhancement**: Enable Stripe Tax and multi-currency dynamic pricing for international hackathon organizers.

4.5 Phase 9: Automated CI/CD Deployment Pipelines (Q3 2027)

- **Current State**: Manual build and deploy scripts (`npm run build`, `npm run server`).
- **Planned Enhancement**: GitHub Actions pipeline triggering automated test suites, Docker container builds, and blue-green deployments on merge to `main`.

---

5. Technical Debt & Known Limitations

Item	Current Implementation	Impact	Recommended Action
Rate Limiter Storage	In-memory via `express-rate-limit`	Limits apply per server pod, not globally across clusters.	Configure `rate-limit-redis` when clustering backend behind load balancer.
Local File Uploads	Stored on local disk directory `uploads/`	Requires persistent container volume mounts in containerized environments.	Map persistent Docker volume or activate S3 driver (Phase 5).
Session Cache	PostgreSQL database table `UserSession`	Increased query traffic on high-frequency requests.	Offload active session token validation to Redis key-value cache.

---

6. Risk Assessment & Operational Mitigations

Identified Risk	Severity	Established Mitigation
Sudden Traffic Spikes During Hackathon Submissions	Medium	Submission deadlines enforce grace periods; client upload chunks files; server utilizes streaming responses.
Database Connection Exhaustion	Low	Prisma connection pooling configured with optimal pool size (`connection_limit=20`).
Credential Loss by Administrators	Low	Multi-factor authentication includes cryptographically generated recovery codes stored in `UserRecoveryCode` .
Tampering with Certificate Credential Claims	Very Low	Certificates feature SHA-256 verification codes verifiable via public verification URL.